Apache OpenOffice

Version 4.1

Draw Guide

AOO Documentation Team

Chapter 6

Editing Pictures (Raster Graphics)

Copyright

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Note for Mac Users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

Windows/Linux	***Mac equivalent***	***Effect***
Tools > Options menu selection	**OpenOffice > Preferences**	Access setup options
Right-click	*Control+click*	Open context menu
Ctrl (Control)	⌘ (Command)	Used with other keys
F5	*Shift+⌘+F5*	Open the Navigator
F11	*⌘+T*	Open Styles & Formatting window

Contents

Introduction

Earlier chapters of the Draw Guide have dealt only with vector graphics, that is, images created by mathematical formulas that specify the relationships between the elements of the image. However, Draw also contains several functions for handling raster graphics (also called bitmaps), photographs, and scanned pictures, as well as imports, exports, and conversions from one format to another.

Draw can read all common graphics file formats. It has some of the same capabilities as specialized programs like Adobe Photoshop or Gimp.

Importing Raster and Vector Graphics

To import stored graphics files, choose **Insert > Picture > From File** on the menu bar or click the  icon on the Draw toolbar. Draw possesses import filters for many different vector and raster graphics formats. If your file has a nonstandard extension, you must choose the format explicitly when importing it.

If you select the **Preview** option in the Insert picture dialog (Figure 1), Draw shows a preview of the picture in the box on the right-hand side. This makes it much easier to choose the picture you want and to see whether Draw can import a file of this format.

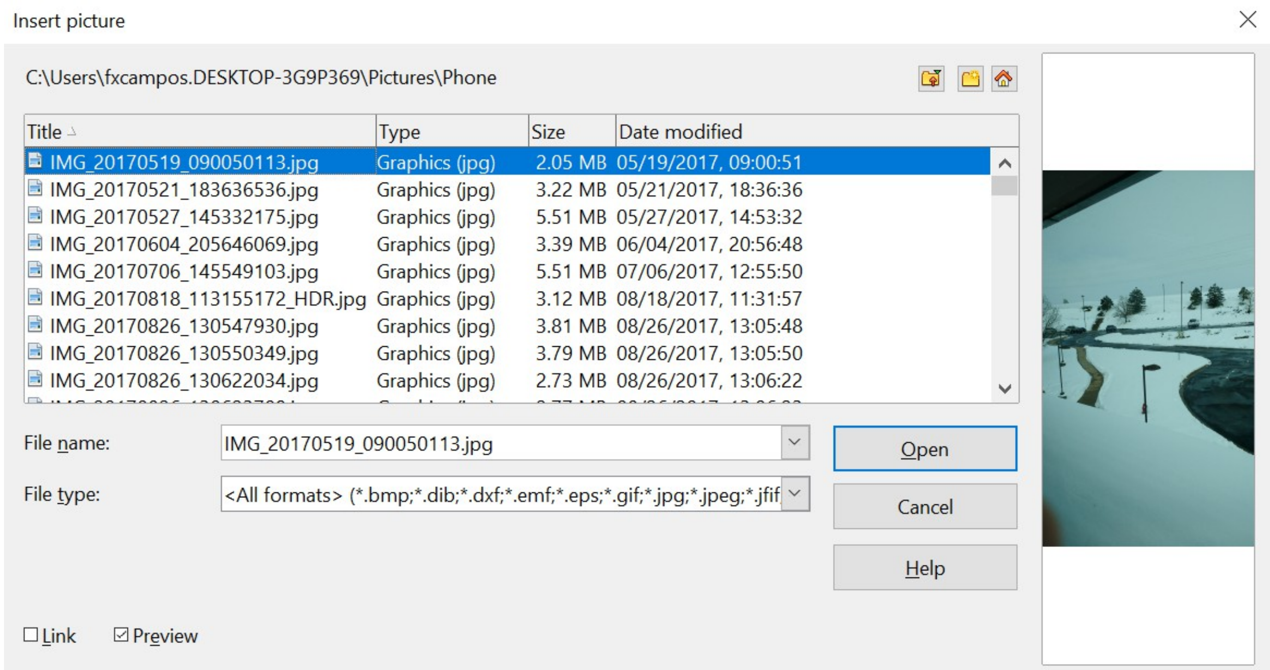


Figure 1: Inserting a picture; notice Link and Preview options in lower left of dialog

Link or Embed?

If you select the **Link** option in the Insert picture dialog, the graphic is linked to, rather than embedded. It's not inserted into the document. Instead, a link to the graphic is created. This link is relative to the folder where the document is stored, even though it shows up in the Edit Links dialog (see Figure 2) as an absolute link.

Note	If you store the document and the graphic in the same folder and transport the folder as a whole to another computer, the graphic will appear in the document as before.

Linked graphics are not changed as a result of any actions performed within OpenOffice. Such changes affect only the view of the graphic in the document, not the graphic itself. In particular, the format of a linked graphic remains unchanged. In contrast, when a raster graphic is embedded in a document, it is converted into PNG format.

Linking keeps the Draw document's file size small. The picture can be edited or even replaced by another, and the link will still function. As long as the new picture is given the same name as the old one, links will be re-established and updated when the document containing the link is next opened. However, some actions do not last beyond the current session (for example, Filter) or are simply not possible on a linked graphic (for example, using the color replacer to exchange colors).

Links can easily be removed. The linked picture will then be embedded in the document. To break a link, choose **Edit > Links** from the main menu bar. In the Edit Links dialog (see Figure 2), choose the link to be broken and then click on the **Break Link** button.

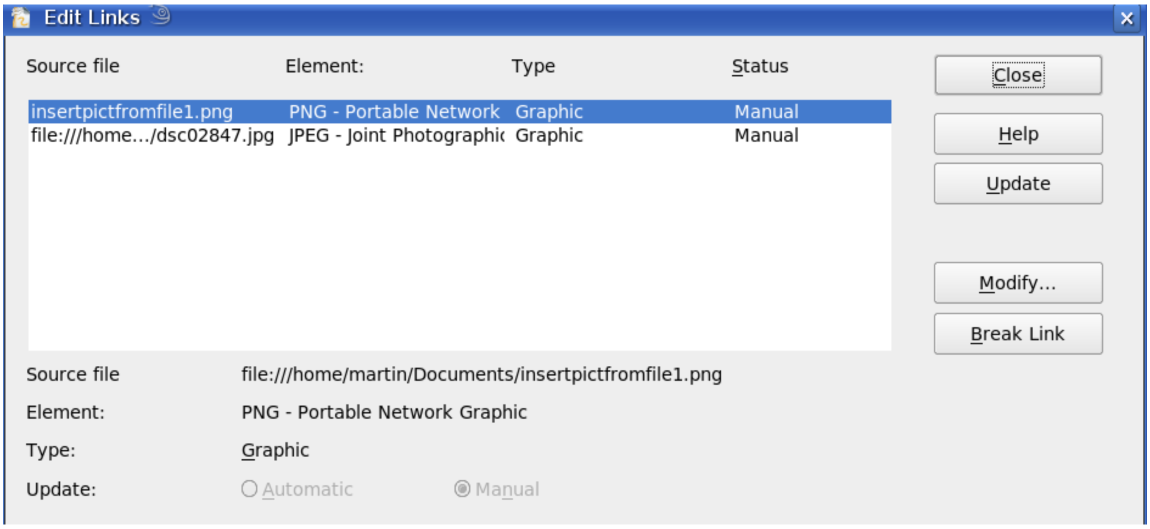


Figure 2: Editing links to documents

Scanning

With most scanners, you can directly insert a scanned picture into a document. Scanned images are embedded in the document in PNG format.

Make sure the scanner is supported by the SANE system on Linux or other UNIX-like operating systems, or by TWAIN if you are using Windows. Note also that either of these must already be configured on the machine running OpenOffice. If more than one scanner or equivalent device is present, you can select the source from **Insert > Picture > Scan > Select Source**.

To insert an image from the scanner:

1) Prepare the image on the scanner and make sure that the scanner is ready.
2) Choose **Insert > Picture > Scan > Request**.

3) The rest of the procedure depends on the scanner driver and interface. You will normally be required to specify the resolution, a scan window, and other options. Consult the scanner’s documentation for more information.
4) When the image is scanned, Draw places it on the page. At this point, it can be edited like any other image.

Pasting from the Clipboard

The clipboard offers another way to insert graphics. Depending on the source and the operating system, figures may be in different formats. You can obtain an overview by choosing **Edit > Paste Special** from the menu bar or by clicking on the **Paste** icon’s drop-down menu on the main toolbar. The choices on the list vary with the type of graphic on the clipboard.

Dragging and Dropping

Drag-and-drop is also possible in many cases. The exact way drag-and-drop works is determined by the operating system in use and the source of the graphic, regardless of whether the graphic is embedded or linked. The behavior can be controlled by pressing the *Control* or *Control+Shift* keys together.

Draw objects and images that are used frequently can be stored in the Gallery. Objects can be dragged from the Gallery to the Draw surface quite simply. Working with the Gallery is dealt with in Chapter 10 (Advanced Draw Techniques).

Inserting from a File

With this option, you can insert complete pages or single objects from existing Draw or Impress documents into your Draw document. Additionally, you can insert text in Rich Text Format (RTF), HTML format, or plain text. The text will be contained within a text frame. The usual paragraph and character formatting options are available for this text.

Choosing **Insert > File** brings up the File selection dialog. Select the required file and click **Insert**. If the file is a Draw or Impress document, the Insert Slides/Objects dialog (Figure 3) opens.

To access single pages or slides of the document, click on the expansion symbol (usually a + or a small triangle, depending on your operating system) to the left of the file name in the selection area. The same method is used to display single objects within slides. Raster graphics (for example, photos) and Metafiles are marked with a , drawing objects with a , and OLE objects with a 
.

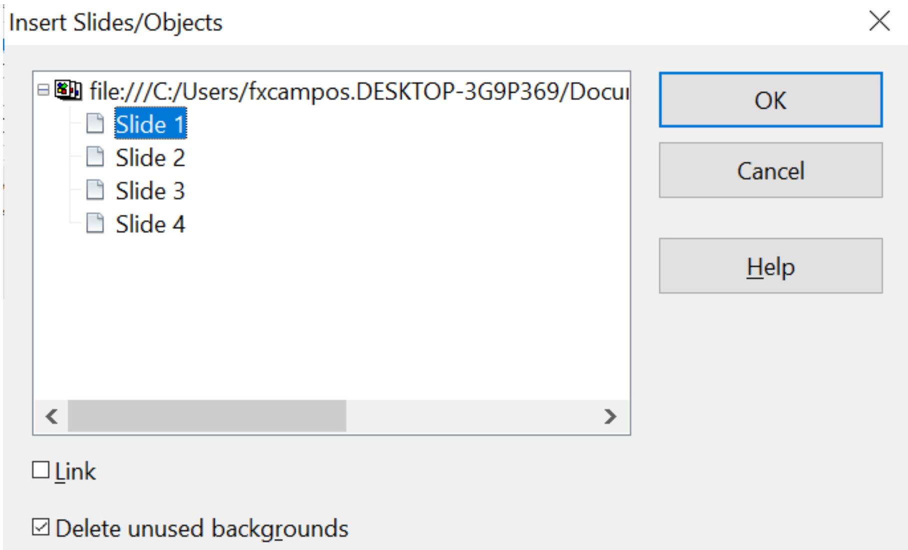


Figure 3: Inserting complete pages or objects on a page

Choose the slides or objects you want to insert; for multiple selections, press the *Control* or *Shift* key while clicking on the items. Then click **OK** to insert the selected items.

If the inserted object was named in the source document, it keeps its original name, unless the name already exists in the current document. In that case, you must give the object a new name before inserting it. To rename an inserted object, right-click and choose *Name* from the pop-up menu. Renaming has the advantage that the object is then listed in the Navigator.

You can choose whether to embed or link the selected page or object.

Exporting Graphics

Draw saves drawings by default in the *.ODG format. Some programs cannot open these files. To use the drawings in other programs, you can export the graphics in various formats. Choose **File > Export**, then, in the File format pull-down list, select the desired format (see Figure 4).

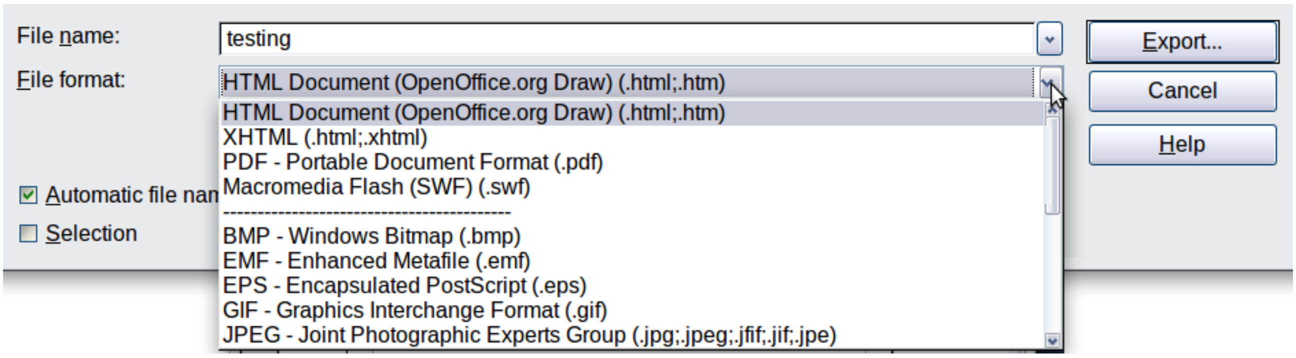


Figure 4: Lower part of the Export dialog

Exporting the Entire File

The file format options in the upper area of the File format box (HTML, XHTML, PDF, and Flash) always apply to the complete file. Using Flash or HTML causes each page of the Draw document to be exported as a raster graphic. After using this option, you will no longer be able to access individual objects on the page.

Flash reached its end-of-life on December 31, 2020. Thereafter, Adobe stopped all support, including security updates. As a result, Adobe strongly recommends uninstalling Flash Player, and major browsers have already removed it.

- **Official End-of-Life:** Adobe officially announced the end of support in 2017, and the final end-of-life date was December 31, 2020.
- **Blocking and Removal:** After that date, Adobe began blocking Flash content from running, and major browser vendors have either disabled or fully removed the Flash Player component from their products.
- **Reasons for deprecation:** The discontinuation was due to several factors, including the maturity of open standards like HTML5, WebGL, and WebAssembly, which offer better performance and security. Flash had persistent security vulnerabilities, and its performance was often poor, consuming high amounts of CPU and battery.

To export a multi-page Draw document as a series of web pages, choose **File > Export** and select **HTML Document** as the file type. The HTML Export wizard opens. Follow the prompts to create the web pages. If you wish, the wizard can generate a navigation aid to help in moving from page to page.

If you want to use the objects in other applications, select one of the Metafile formats (JPEG, PNG, BMP, and so on) in the lower part of the File format list (see Figure 5). With this type of export, only the currently active page is exported.

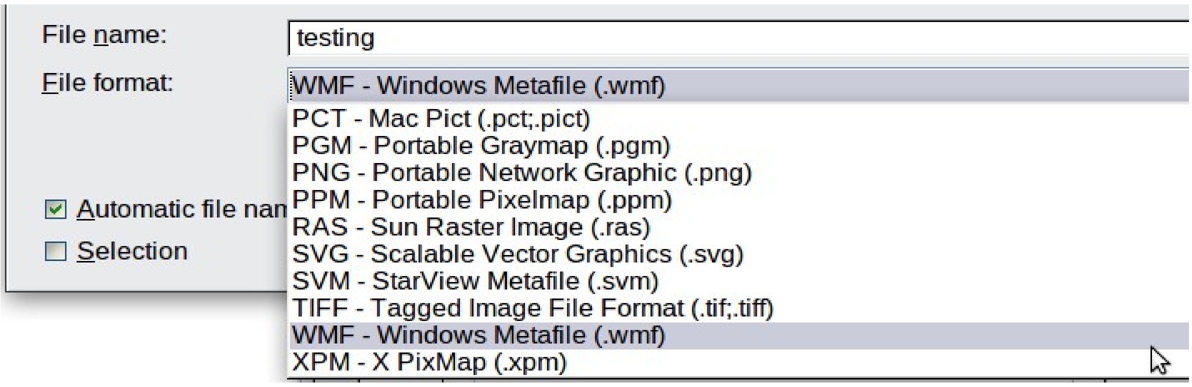


Figure 5: Section of the file selection list

Exporting Single Objects

To export individual drawing objects, you first need to select them. A selection can include more than one object. Take care to choose the *Selection* option from the Export dialog.

Exporting as a Vector Graphic

The choice of vector formats is still limited. Even exporting a graphic to an SVG file is not yet fully implemented: 2D graphic objects are exported as paths, and 3D objects only as preview images. The Metafile formats are best supported; exporting to one of them is usually successful.

Exporting as a Raster Graphic

The choice of export formats for raster graphics is large enough to support most other applications. In case of doubt, test different formats to see which gives the best results; some programs behave differently depending on the image formats they import.

After entering a name for the exported file and selecting the file format, you may be able to set options for the exported image—compression, color format, and version of Metafile—depending on the format you choose. Some examples of the various option dialogs are shown in Figure 6.

Where the export dialog allows you to specify the image resolution, this does not affect the number of raster points in the image. Rather, it inserts information into the picture that tells other programs the dimensions to use when displaying the image. This means that, among other things, missing raster points can be introduced when the graphic is opened in another program.

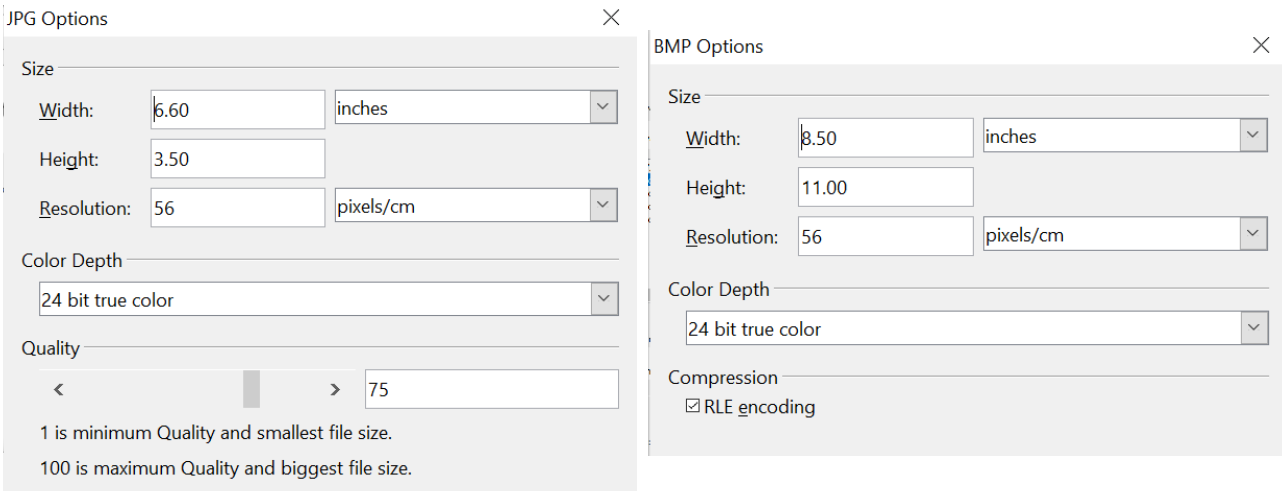


Figure 6: Adjusting format specific properties prior to export

The actual number of pixels used is determined by the screen resolution set up by the operating system, and the Drawing scale factor set in **Tools > Options > OpenOffice Draw > General.**

Embedded raster graphics, or objects that have been converted to a bitmap or Metafile (from the context menu **Convert > To Bitmap / To Metafile**), may subsequently be saved as a picture; from the context menu, choose **Save as Picture**. This saves the picture without Draw-specific additions such as text elements, borders, or shadows, and without the possibility of setting any of the options available when exporting.

Draw objects can easily be imported into Writer, Calc, or Impress documents. For objects used often, this is most easily done by storing them in the Gallery or using the clipboard. Writer and Calc do not possess the same range of tools as Draw. So it makes sense to use Draw to produce a complex drawing, and then to copy it into other AOO applications. A link to single drawing objects is not possible, but you can incorporate Draw documents in the other modules as linked OLE objects.

Tip	To avoid problems when scaling Draw objects containing text after importing them into Writer, convert the object to a polygon before storing it in the Gallery or copying it to the clipboard. See "Convert to a Polygon" on page 21.

Modifying Raster Object Properties

As with other objects, the properties of a raster graphic can be modified. You can format the graphic using the Graphic pane of the Sidebar, the Format menu, or the context menu. Use the Picture toolbar to add or change filters using the Graphic Filter Bar; adjust the *Lines*, *Areas*, and *Shadows* properties. The *Transparency* property in the Format menu does not relate to the transparency of the raster graphic itself but to the background area. To set the transparency of the graphic, you must use the Picture toolbar.

Graphics can also have a text element. For more on text, see Chapter 10 (Advanced Draw Techniques).

You can change the position and size of graphics, or rotate them, using the methods described in Chapter 3 (Working with Objects and Object Points). Raster graphics can be flipped (**Modify > Flip**), but note that some Metafile formats may have problems flipping text.

Graphics included as a member of a group behave like other drawing objects when the properties of the group are modified.

It is a good idea to (re-)name the graphic using **Name** from the **Modify** menu or the context menu. Only named objects are visible in the Navigator, and only named objects can be directly imported from another file.

The Picture (Editing) Toolbar and Graphic Pane

If you have enabled the Picture toolbar (in **View > Toolbars**), it will automatically appear on the screen whenever you select a (bitmap) picture (see Figure 7). It appears either directly under the menu bar in place of the formatting toolbar or as a floating toolbar elsewhere on the screen.

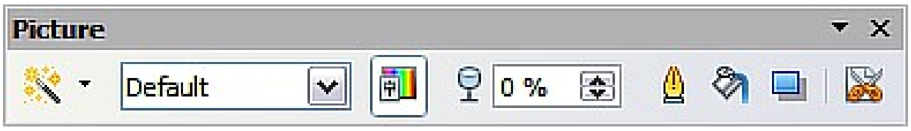


Figure 7: The Picture toolbar

Most of the same tools are available on the Graphics pane of the Properties deck of the Sidebar. A Graphics pane, like that in Figure 8, appears automatically when you select a picture.

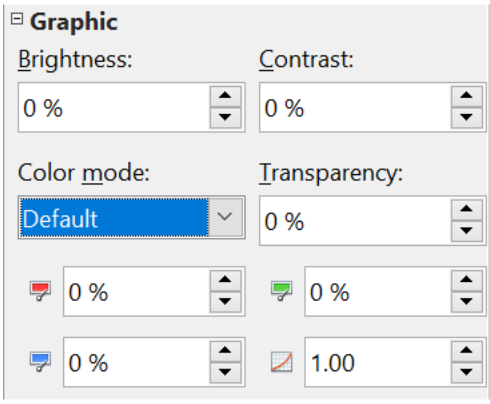
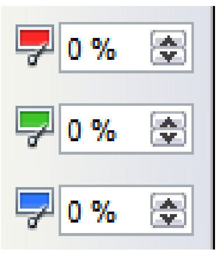
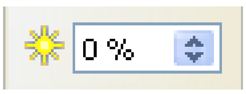
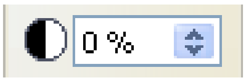


Figure 8: The Graphic pane on the Sidebar

The following table explains the individual functions on the Picture toolbar and illustrates their use with examples.

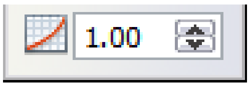
Filter 	Opens the Filter toolbar, which is described in "The Graphic Filter Toolbar" on page 14. This tool is not on the Sidebar.
Graphics Mode 	Use the Graphics mode (*Color mode* on the Sidebar) menu to change the display of the graphic from normal color to grayscale, black and white, or a watermark. This setting affects only the display and printing of the picture; the picture itself remains unchanged.
Default:	The graphic is displayed unaltered in color.
Grayscale:	The graphic is displayed in 256 shades of gray.
Black/White:	The graphic is displayed in black and white.
Watermark:	The brightness and contrast of the graphic are reduced to the extent that the graphic can be used as a watermark (background).
Color 	Use the Color tool to adjust the values of the three RGB colors, the brightness, contrast, and the Gamma value. These adjustments do not affect the original picture, but the values are stored in Draw as a separate formatting set.
	You can select values from between –100% (no color) to +100% (full intensity); 0% represents the original value of the property.
Brightness 	The brightness can be adjusted between –100% (totally black) and +100% (totally white).

Contrast



The contrast can be adjusted between –100% (minimum) and +100% (maximum).

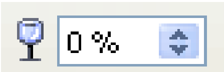
Gamma



The Gamma value affects the brightness of the middle color tones. Values can range from 0.10 (minimum) to 10 (maximum)

Hint: Try adjusting this value if changing brightness /contrast does not give you the result you want.

Transparency



The degree of transparency of the picture can be adjusted from 0% (opaque) to 100% (fully transparent).

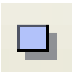
Line

Opens the Line dialog. In this context, line refers to the outline of the border. See also Chapter 4 (Changing Object Attributes). A subset of the tools are on the Line pane of the Sidebar and the full menu can be accessed via the three dots on that pane.

Area

Opens the Area dialog. Here you can edit color, gradient, hatching, and fills of the background area that contains the graphic—not the graphic itself. To see the background, you must set the graphic's transparency to a suitably high value. A subset of the tools are in the Area pane of the Sidebar, and the full menu can be accessed via the three dots on that pane.

Shadow



Use this tool to set a default shadow effect around the graphic. More control of the shadow is available on the Area dialog.

Crop

Use this function to crop (trim) an image. When you click on this icon, crop marks appear on the image (Figure 9). Drag one or more of these marks to crop the picture to your desired size. For more accurate cropping, see “Cropping” below. This tool is not available on the Sidebar.

Figure 9: Interactive crop marks on an image

Cropping

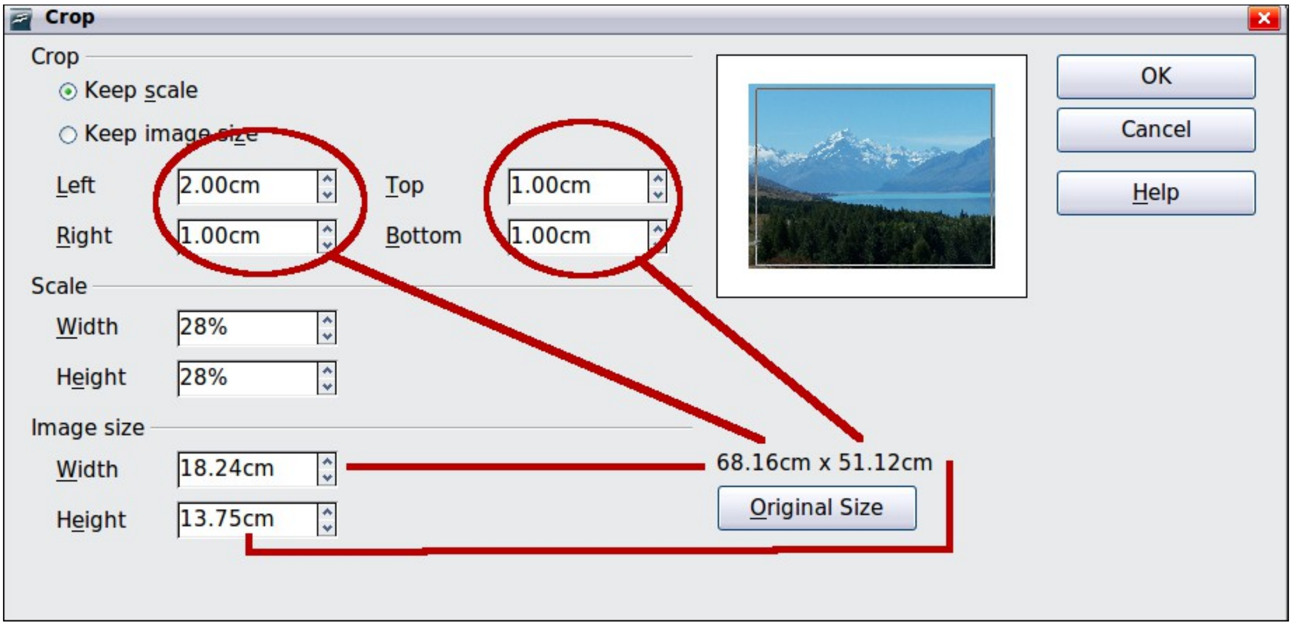


Figure 10: The Crop dialog

More control and accuracy over cropping functions are available via the Format menu. Click **Format > Crop Picture**, and the dialog box in Figure 10 will appear. The fields *Left*, *Right*, *Top*, and *Bottom* specify the amount to trim from the appropriate edge.

In addition to cropping, you can enlarge or reduce the graphic's size by changing the percentage scaling values. The new dimensions of the graphic are shown in the *Image Size* width and height boxes, which can also be directly adjusted.

In Figure 10, you can see that 2 cm from the left side and 1 cm from the top, bottom, and right side will be trimmed from the original object. The preview pane shows the location of the new edges of the graphic. In addition, both height and width have been scaled down by 28%.

- If you choose *Keep Scale*, the graphic will be cropped to the scales shown in the width and height boxes, and the picture will be reduced in size accordingly.
- If you choose *Keep image size,* the graphic will be cropped and then enlarged to the original image size.

Take care with these operations; in the Crop dialog, the width and height are treated as totally independent values. Changing one without the other can result in significant image distortion. In that vein, be aware that the Position and Size dialog in the context menu has an option to keep the width-to-height ratio fixed while changing one of the two dimensions. Changing values in one area (Scale or Image Size) will be reflected in the other area.

Caution	Changes made in the Crop dialog affect only the view of the picture. The original picture is not changed. To export a cropped graphic, you must use **File > Export**. If you use the option **Save as Picture** from the context menu, the changes are not exported.

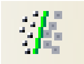
The Graphic Filter Toolbar

Click the **Filter** icon to open the Filter toolbar. Draw offers eleven filter effects. Filters work on the current view of an object, and can be combined. Filters always apply to the entire graphic; it is not possible to use filters to edit only a part of the object.

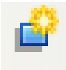
Invert reverses (inverts) the colors of an image so that it appears as a color negative of the image.

Smooth reduces the contrast between neighboring pixels, resulting in a slight loss of sharpness. If you use the filter several times in a row, the effect will be strengthened.

Sharpen increases the contrast between neighboring pixels, emphasizing the brightness difference. This will accentuate the outlines. The effect will be strengthened if you apply the filter several times in a row.



Remove noise compares every pixel with its neighbors and replaces the extreme values (those that deviate in color by a large amount from a mean value) by a pixel with a mean color value. The amount of picture information does not increase, but because there are fewer contrast changes, our brains can better recognize the resulting graphic. This filter tends to make the picture a little smoother.

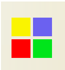


Solarization was originally a photochemical effect. If the location of a photograph is extremely highly lit, you can experience a reversal of color and brightness. Similarly, entry of light during the developing process reverses the brightness values. These phenomena were used for artistic production of pictures.

With this filter, you input a threshold brightness value, above which the color values are reversed (middle picture, threshold value 70%). With the Invert option, the whole of the resulting picture is inverted in color (right picture).

Aging lends pictures a certain “look” resembling that of old photos. The process first produces a grayscale picture from the original and then reduces the intensity of the blue and green color values, so that the final picture appears darker and redder. In the middle example, the aging degree was set to 0%, on the right it was 15%.

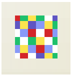
Posterize reduces the number of colors in the picture. The fewer colors, the flatter the picture appears. In the picture below on the right, the number of colors was reduced to 8. The results of this filter are not always good.



The **Pop-Art** filter is supposed to change the colors of the picture to a pop-art format, but unfortunately, it does not seem to function correctly at present.

Charcoal sketch makes the picture appear as if it had been drawn with charcoal. The outlines are in black, and the original colors are suppressed.

The **Relief** filter calculates the edges in relief of the picture and produces a picture as if illuminated by a light source, the position of the illuminating light being variable and producing shadows that differ in direction and magnitude.



Mosaic takes groups of pixels and converts them into a single color rectangular tile. The whole picture appears to be a mosaic. The center and right pictures below had an element resolution of 5 pixels. The picture on the right also had the *Enhance edges* option selected; with the greater contrast at the edges, it appears to be a little sharper.

Caution	If your picture is linked, filters are applied only to the current view. The stored picture is not changed. When you close the document, all filtering is lost. You should ensure that you export the picture to create a copy with all the filters applied (**File > Export**). If you embed the graphic in the document, all filters are applied directly to the embedded graphic and cannot be undone in a subsequent session. If you do not want to retain a filter, you must use **Edit > Undo** to return to an earlier state of editing. After you save and close the document, the filter effects are permanent.

Changing Colors Using the Color Replacer

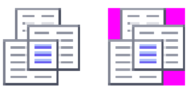
Use the Color Replacer to change one color in a picture with another, or set the color to transparent. The tool always works on the entire picture; you cannot select only a region for editing.

The changes are applied to the graphic itself and therefore require it to be embedded. Attempting to use the eyedropper tool on a linked graphic will result in the following message: *This graphic is linked to a document. Do you want to unlink the graphic in order to edit it?*

The tool can be used on all raster graphic formats and with many—but not all—Metafile formats.

Practical Example: Changing Wrong Colors to Transparency

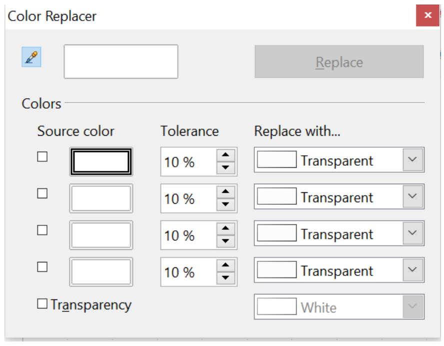
Some applications that cannot correctly handle transparency show transparent areas with the color Magenta, then store the graphic with this color and without the correct transparency information.



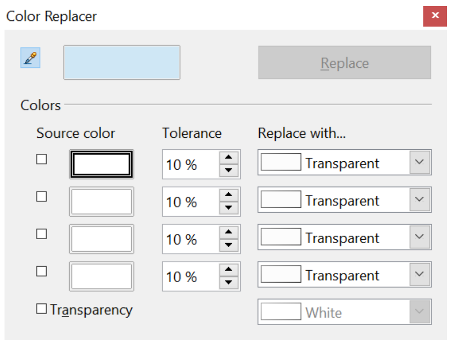
Bitmap graphic with area in magenta, which actually should be shown transparent. On the left is the original picture, right is the picture after saving it with MS Paint.

If you receive such a picture, you can recreate the transparency with the color replacer tool.

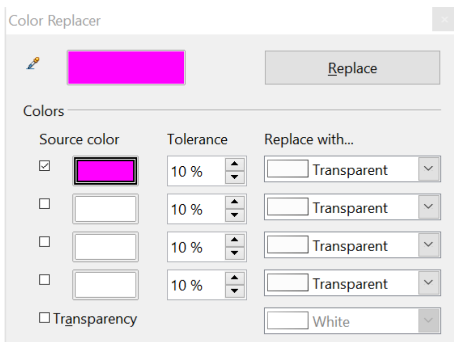
Open the Eyedropper dialog via **Tools > Color Replacer**, then click the picture to be edited.



Click the icon to change to the color selection mode. Outside the dialog, the cursor changes to a hand. The field next to the icon shows the color immediately under the hand cursor.



Click on the color to be changed. The first Source Color box is now marked, and the color selected appears in the left box. In the *Replace with* selection list, the option *Transparent* is already selected. Click the **Replace** button in the upper-right of the box to apply the changes to the picture. There is no preview of the effect. If the result is not what you wanted, choose **Edit > Undo**.



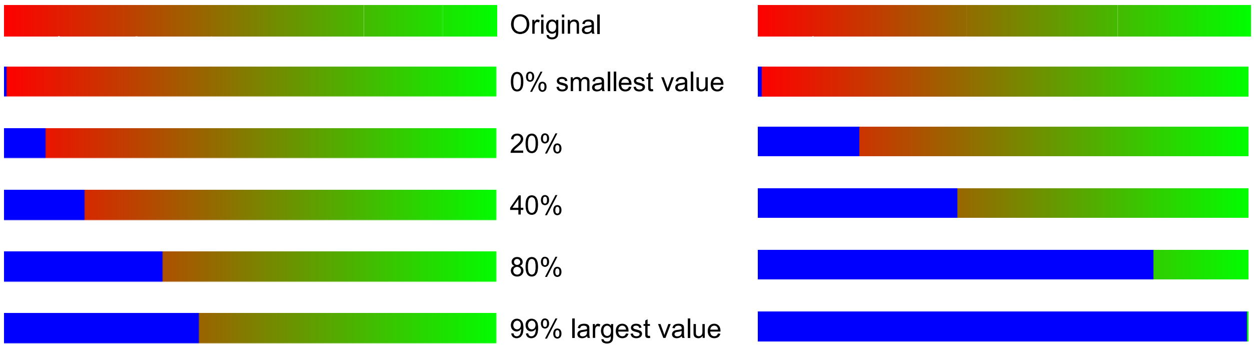
This dialog does not close automatically, so you can carry out further color replacements. First, mark the Source color field. Then choose the replacement color with the cursor. Close the dialog with *Ctrl+F4* or the **Close** icon on the window border. Draw uses a separate Alpha channel for transparency so that pixels of different colors can be made transparent.

It is also possible to perform the operation in reverse, changing a transparent area to a color. Select nothing in the graphic, but choose the Transparency option at the lower left of the dialog and the replacement color at the lower right.

The selection list for replacement colors shows all available colors in the current document's color palette. You cannot define any new colors here, but you can add colors to the available palette before using the color replacer tool. For more on this topic, see Chapter 8 (Tips and Tricks).

Effect of the Tolerance Parameter

The tolerance parameter describes how closely a color value in the picture must match the source color if it's to be replaced. In the figure below, the red color tone at the left edge of the rectangular color palette is used as the source color, and a pure blue as the replacement color, with several different tolerances. The original is a rectangle with a color palette from red to green. Because the color replacer cannot be used on drawings, the rectangle on the left was converted to a raster graphic (**Convert > Bitmap**). The picture was then exported in an Enhanced Metafile format and inserted on the right for comparison.

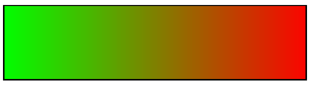


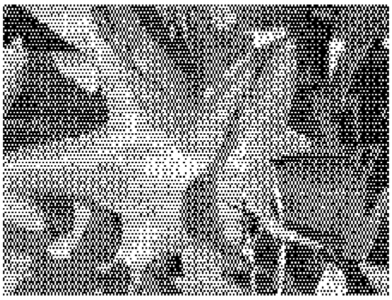
Color Depth

It is possible to reduce the color depth in OpenOffice. The tools are available, but you must create a separate menu and add them to it if you wish to have them available (refer to the help if you are not sure how to do this). To find the tools, click on **Tools > Customize >**, **Toolbars**, and **> Add** (Command). The commands are in the *Modify* category, at the top of the list.

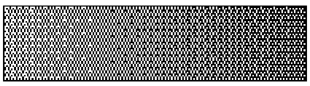
Reducing color depth

Original with a 24-bit pixel depth (=8 bits per channel)





1-bit dithered

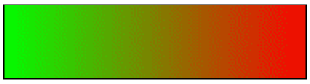


The impression of grayscale is produced by a raster. In reality, there are only two colors.

1-bit threshold

The threshold determines which pixels are black and which are white. You cannot set it directly, but you can influence which part of the image is set to black by varying color settings on the picture toolbar—red, green, and blue level, brightness, contrast, and/or gamma value—to see what works best for your image. You must export and save the modified image with **File > Export** and reopen it, before changing the color depth.

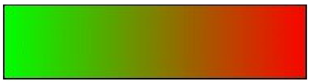
4-bit color palette



With 4 bits, a total of 16 colors can be produced. The color steps are quite fine because the process does not use the 16 basic RGB colors but instead selects from the palette those that best match the colors in the image.

4-bit grayscale palette

8-bit color palette



With 8 bits, a total of 256 colors can be produced. The image on left is, at first glance, very difficult to distinguish from the original. A big difference is that the picture file is only one-third of the size of the 24-bit version.

8-bit grayscale palette

Conversion

Convert to a Contoured Image

Select the picture so you see the green handles. From the context menu or the **Modify** menu, choose **Convert > to Contour**. This command creates a polygon or group of polygons from the original image, with four corner points, and with the image set as a background graphic. In this state, you cannot edit the graphic further. All of your modifications must be completed before this point.

The polygon is, in fact, a vector graphic, but the picture remains a bitmap image. Using a polygon offers possibilities for further change; for example, you can modify the shape or define a transparency gradient.

Convert to a Polygon

Select the picture so you see the green handles. From the context menu or from the **Modify** menu, choose **Convert > to Polygon**,as illustrated in Figure 11. This command converts areas of the same color into small filled polygons. The entire image becomes a vector graphic and can be resized without loss of image quality or distortion of text. The resulting format will be a Metafile.

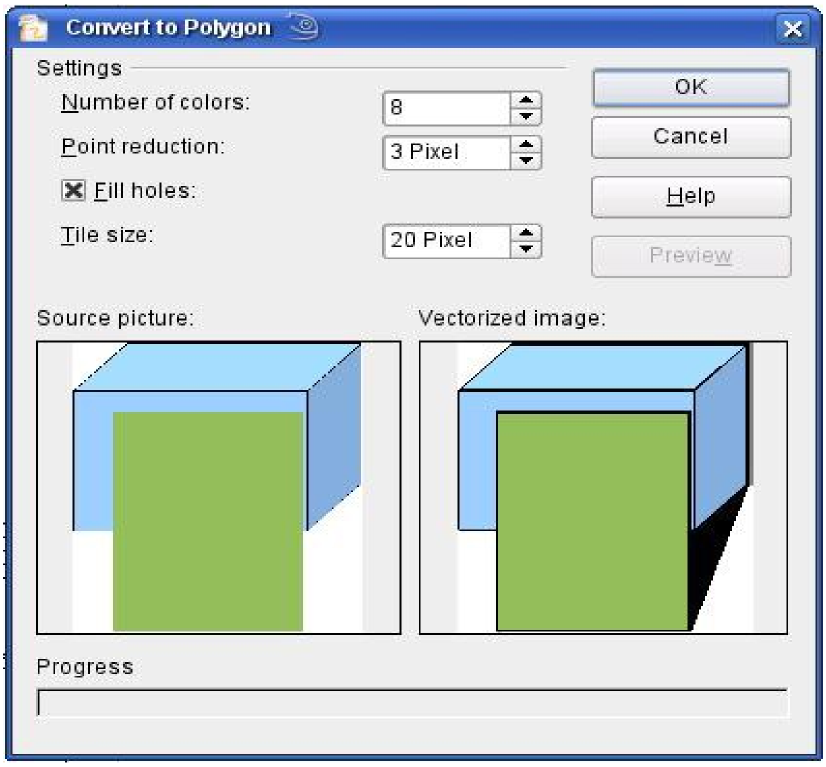


Figure 11: Conversion of a picture to polygons

The Convert to Polygon dialog has several settings that can be tested using the *Preview* button. Because graphics conversion can sometimes take a while, depending on both the picture and the computer, a progress bar is included at the bottom of the dialog.

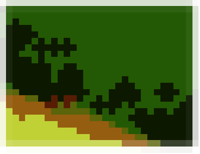
Number of Colors

Draw considers 8 to 32 colors during conversion. However, the picture can contain fewer than 8 colors. For every color in the picture, Draw creates a polygon that may consist of several disconnected parts. These polygons are then filled with the relevant color.

The algorithm Draw uses to reduce the number of colors in the picture to the number specified is not yet optimal. You may prefer to reduce the number of colors by changing the color depth or using the *Posterize* filter (page 16) from the Graphic Filter toolbar (page 14).

Point Reduction

Polygons will only be created if their size is greater than the value given in this setting. The section of the picture below, highly magnified, shows that with a larger point count, small flecks of color—typical of irregular color patterns—simply disappear.

2 Pixel  3 Pixel 

Fill Holes

Point reduction can result in small areas, or “holes,” that are not covered by a polygon. If the **Fill holes** option is marked, additional square, tiled areas are created with a background color matching the hole. The **Tile size** option allows the width in pixels of these square areas to be preset.

Original picture

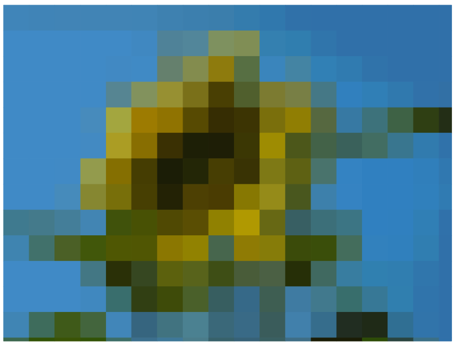
Original picture with 16 colors and 0-pixel-point reduction vectorized

Original picture transformed with the Poster filter and 64 colors …

… and then with 16 colors and 0-pixel-point reduction vectorized.

Here, the picture is transformed with the Poster filter, but this time with a point reduction of 4 pixels and a tile size of 16 pixels.

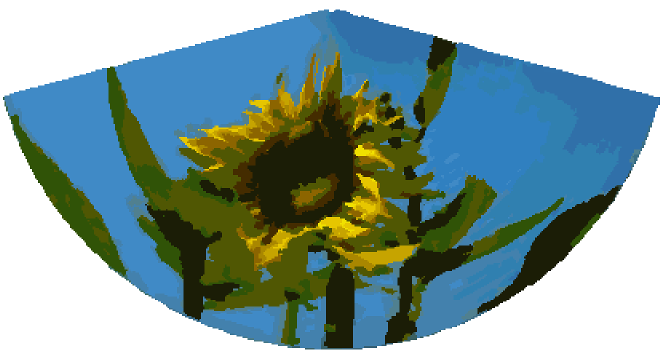
The effect below was generated from the picture to the left (posterized and vectorized); the polygons were split with **Modify > Break**, and a number (in this case 6) of the (foreground) polygons were deleted until the resulting background was achieved.



Note: To select polygons for deletion, click on a curved colored area (the text in the status bar shows “Polygon nnn corners selected”, where nnn is the number of corners of the polygon) and a frame with green handles encloses the area of the polygons selected. Press *Delete* to delete the polygons. Repeat until you have achieved the desired effect (which here is to have only the background showing).

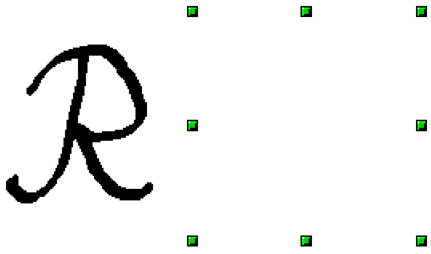
If you next ‘split’ the existing Metafile (**Modify > Split**), access is gained to individual polygons. To be able to operate on these together, you should immediately group them after splitting.

With the vectorized form of an image, you can carry out operations familiar to those from classic drawing programs. One example might be a curved warping operation. For such transformations, it is much less effort to use a conversion with background tiles so that the picture's edges are straight.



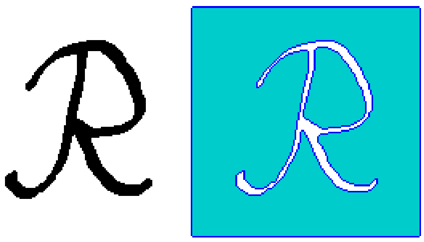
Using the graphic generated in the posterize and vectorize process above, Break and Split it, and then try out some operations using the tools on the Effects toolbar.

To use this tool to vectorize the first letter of a chapter in a special font, convert the graphic to a polygon using no point reduction or background tiling. Use **Modify > Break** to generate two polygons. An example is shown below.



Original polygon, converted from a Metafile. Break the graphic into two polygons.

After the conversion, there are two polygons: one with the visible letter and another, not visible because the lines and area fill are all white. You can select this “invisible” polygon by pressing *Tab* (click the visible letter first, then press *Tab*; this moves the current selection to the next object). In the example at the left, the invisible polygon has been moved to the side. If you have Display Grid active and the grid points set to display behind objects, you can see where this invisible polygon is, since it obscures the grid points behind it.



This invisible polygon encloses the letter. After changing the line(s) and/or area to a different color, the polygon becomes visible.

Convert to Bitmap

All drawing objects are vector graphics. Use the command **Convert > To Bitmap** from the context menu to convert a vector graphic to a raster graphic. Draw creates a raster graphic in PNG format with a 24-bit color depth. Unfortunately, any transparency effects created in the vector graphic are lost during this conversion, although the PNG format used in Draw does support transparency. Only if you use the color replacer tool to set transparency will an Alpha channel also be produced.

Note	An alpha channel represents the transparency of a computer-generated image.

To determine the height and width of the converted graphic, that is, the number of pixels needed, Draw considers the dots per inch or dpi setting of a given operating system for a monitor's screen, as well as the percentage scaling factor set under **Tools > Options > OpenOffice > View** in the section *User Interface*.

Example:

Width of vector graphic: 1.5 inch
Monitor setting of operating system: 96 dpi (= 96 dots per inch)
Scaling 130 %

Calculation: 1.5 inch x 96 dpi *130% = 187 pixels

The actual number of pixels will vary somewhat from this value, due to rounding. When the format permits, Draw uses a dpi value (in this case 125 dpi) which permits the calculation to be reversed, causing a picture with this pixel value to be reproduced with the correct width of 1.5 inches.

Keep in mind that the above example is for displaying a drawing on a screen. Printed images typically require a higher dpi value.

Print Options with Raster Graphics

If you have only a black-and-white printer or are experiencing printing problems, consider the following settings.

You can set up printing so that all text and all graphics are printed in grayscale, that is, black and white. The general settings are in **Tools > Options > OpenOffice Draw > Print**, in the section *Quality* (see Figure 12). To set these for the current document only, use *Options* in the Print dialog (**File > Print**). With both Black/white printing and Grayscale printing, no background is printed.

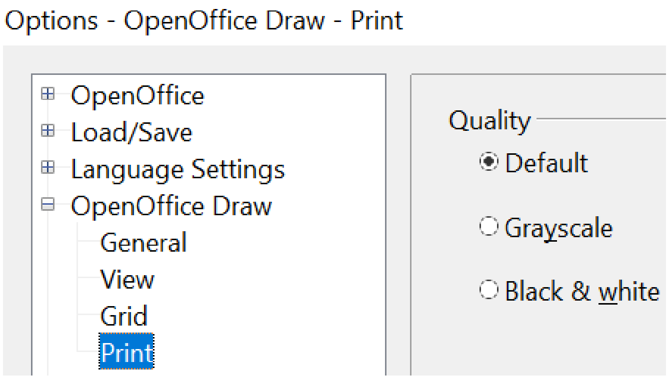


Figure 12: Options - OpenOffice Draw - Print

Other options are in **Tools > Options > OpenOffice > Print** (see Figure 13). They relate to page options (scaling) and other printing variables. They affect the size of the print file and the time taken to print the document. With most modern systems, you can simply accept the defaults. However, because operating systems and printers have unique requirements, you may need to tailor printing settings to your site's conditions. Refer to the Help file for more information.

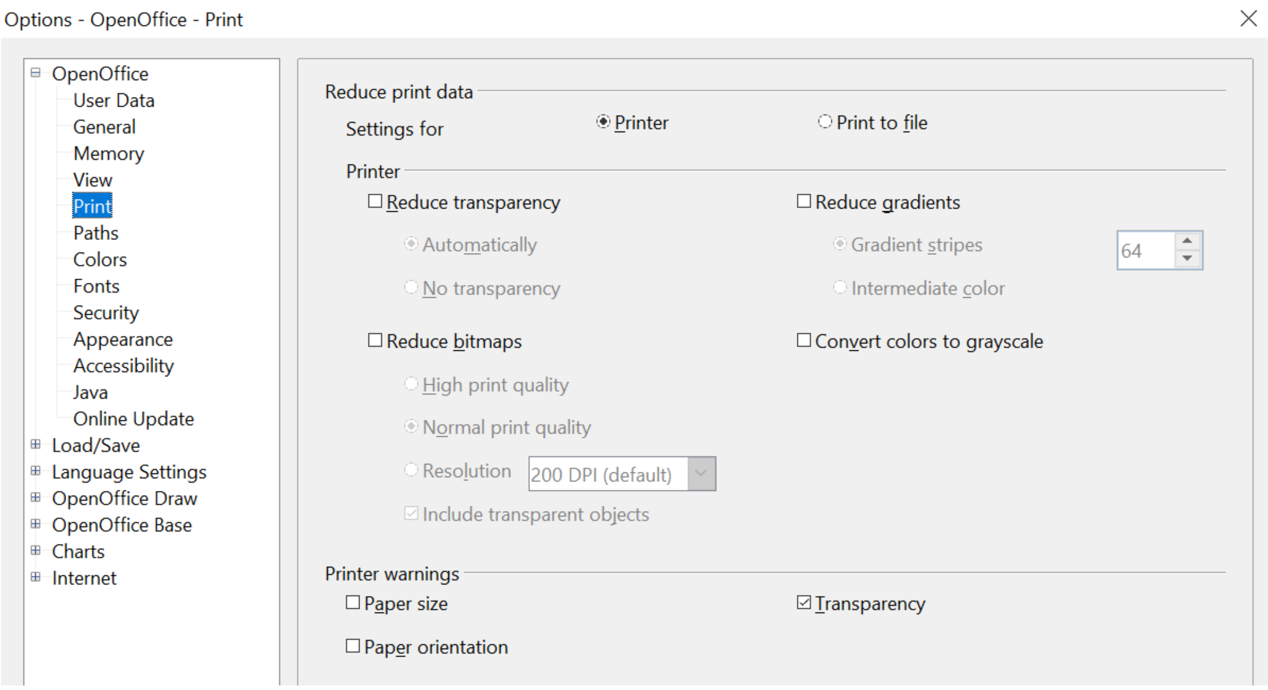


Figure 13: Options - OpenOffice - Print dialog